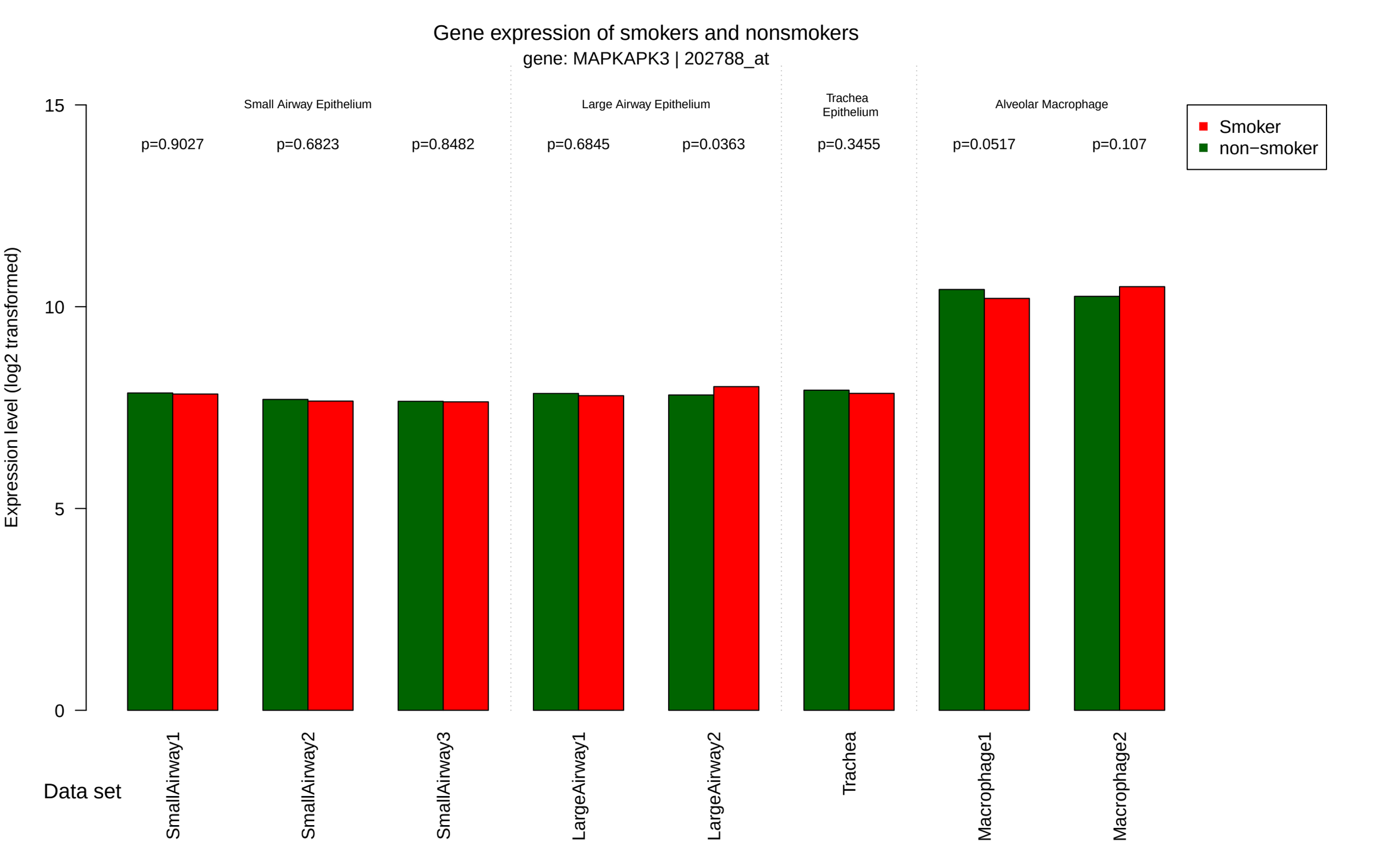
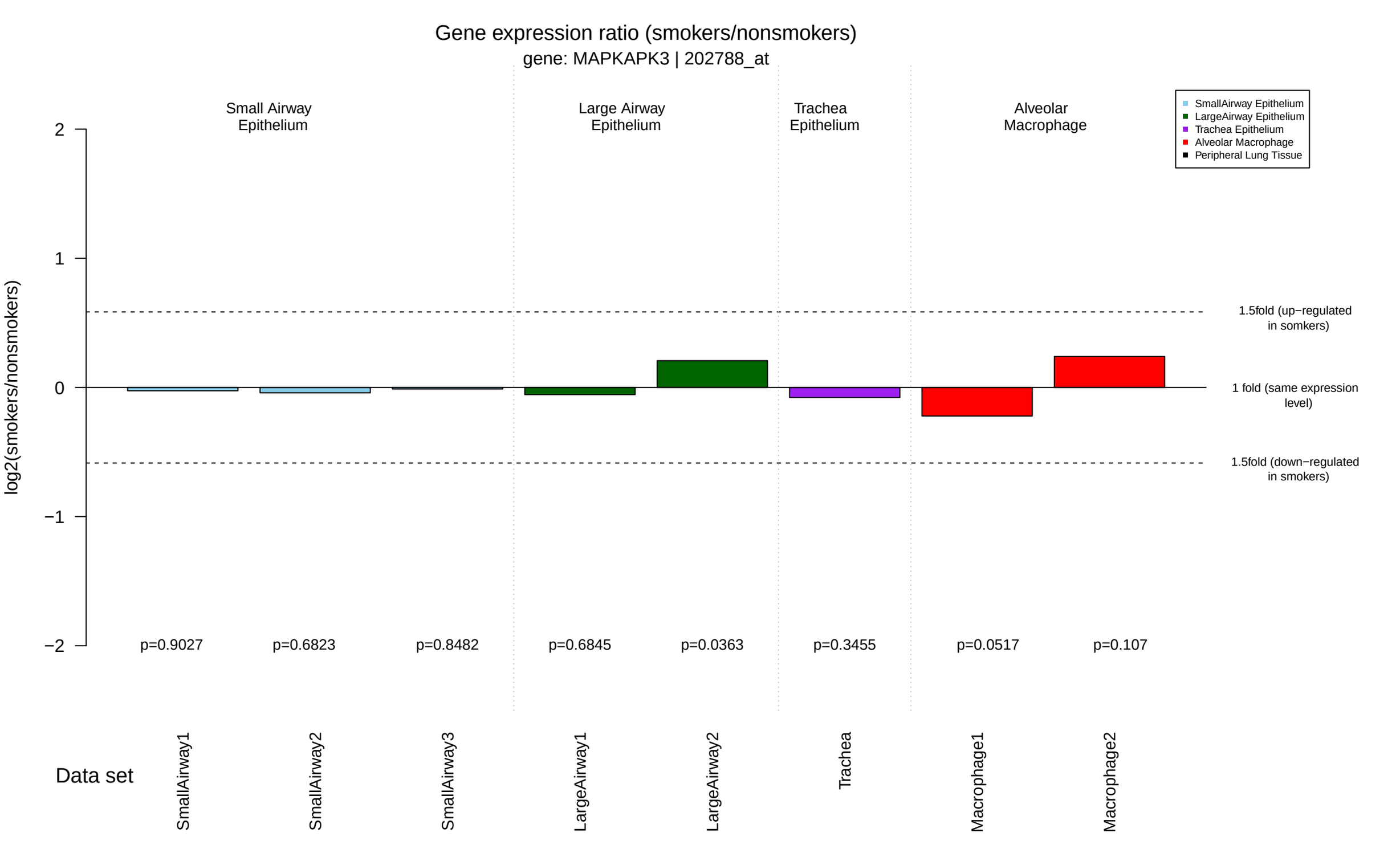
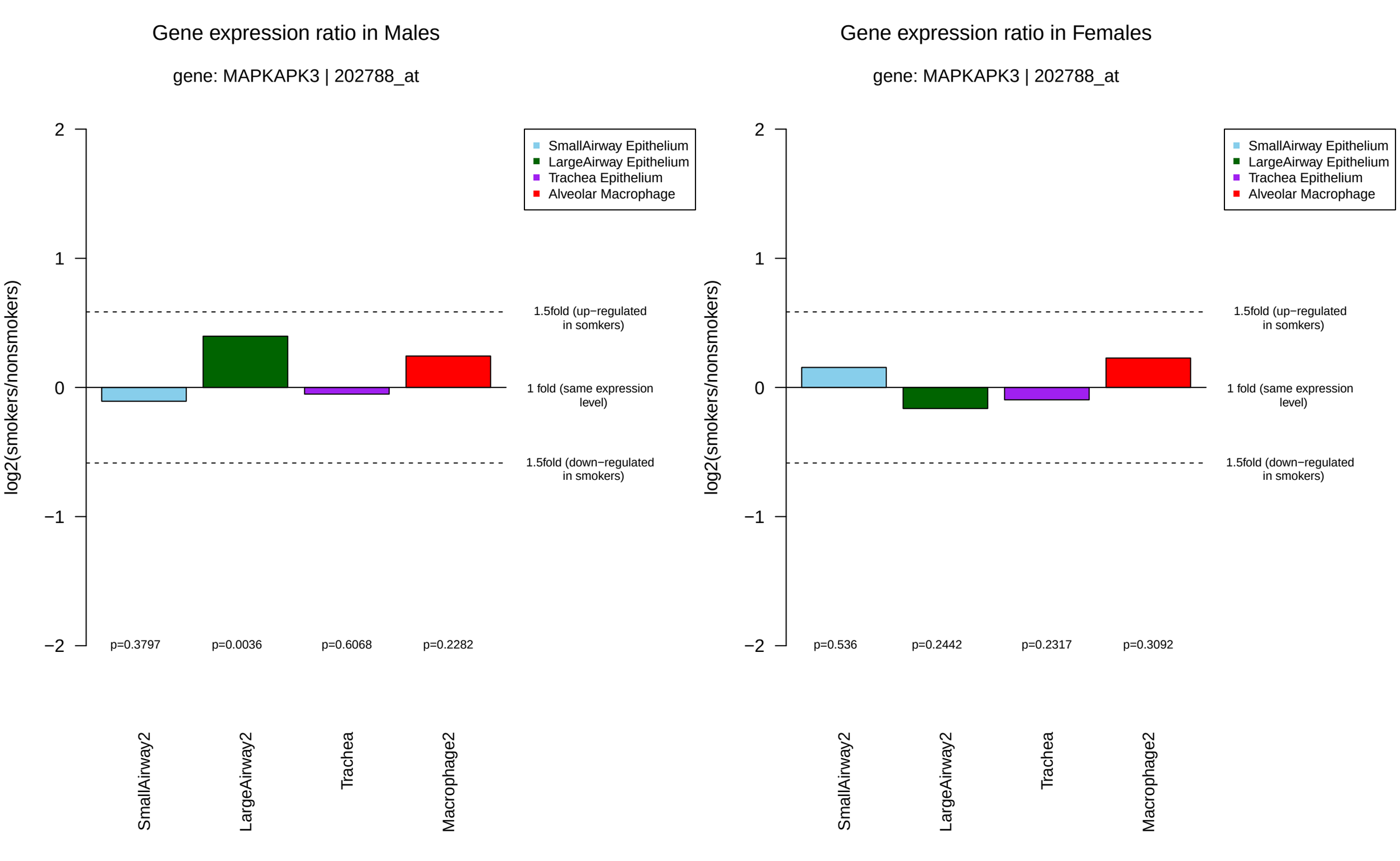
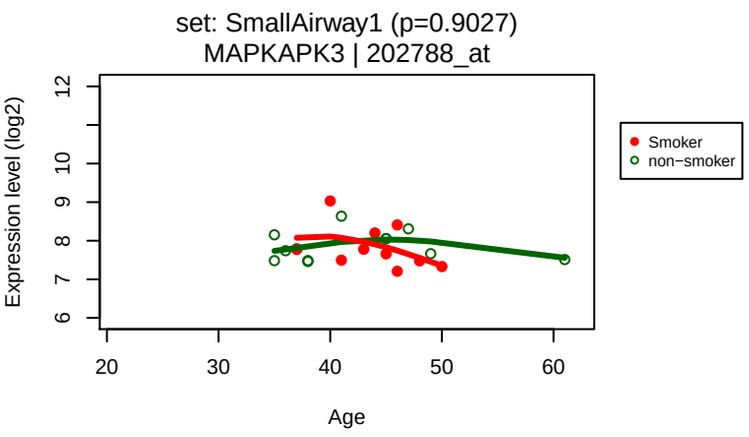
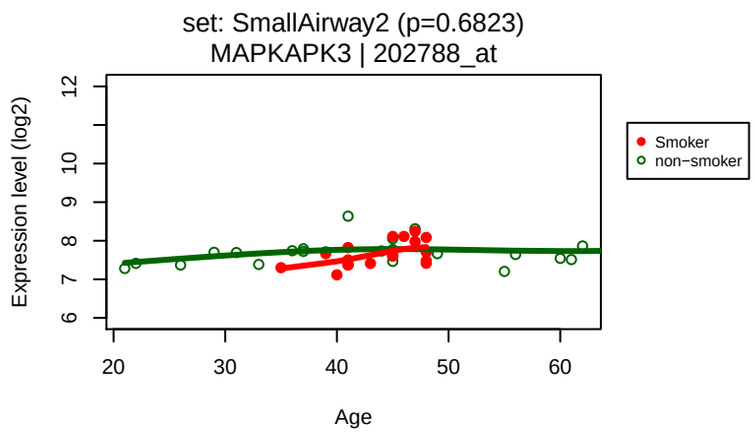
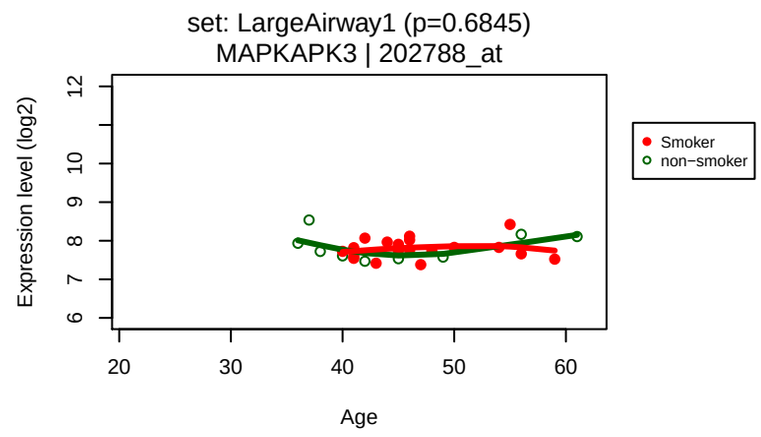

Gene expression of smokers and nonsmokers
gene: MAPKAPK3 | 202788_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9027
p=0.6823
p=0.8482
p=0.6845
p=0.0363
p=0.3455
p=0.0517
p=0.107
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

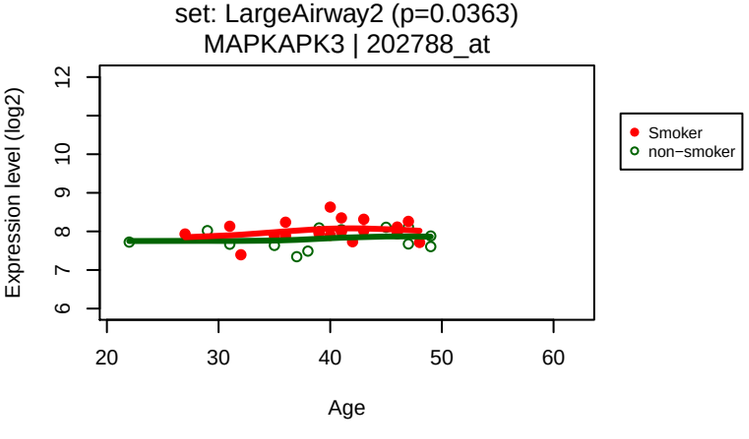

Gene expression ratio (smokers/nonsmokers)
gene: MAPKAPK3 | 202788_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9027
p=0.6823
p=0.8482
p=0.6845
p=0.0363
p=0.3455
p=0.0517
p=0.107
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

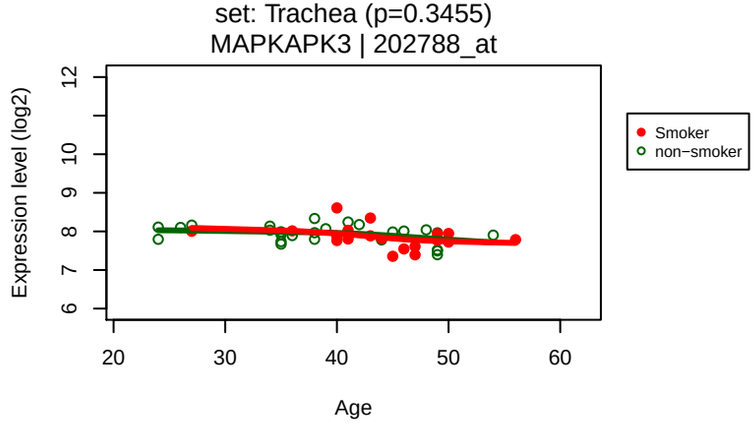

Gene expression ratio in Males
gene: MAPKAPK3 | 202788_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3797
p=0.0036
p=0.6068
p=0.2282
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPKAPK3 | 202788_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.536
p=0.2442
p=0.2317
p=0.3092
SmallAirway2
LargeAirway2
Trachea
Macrophage2

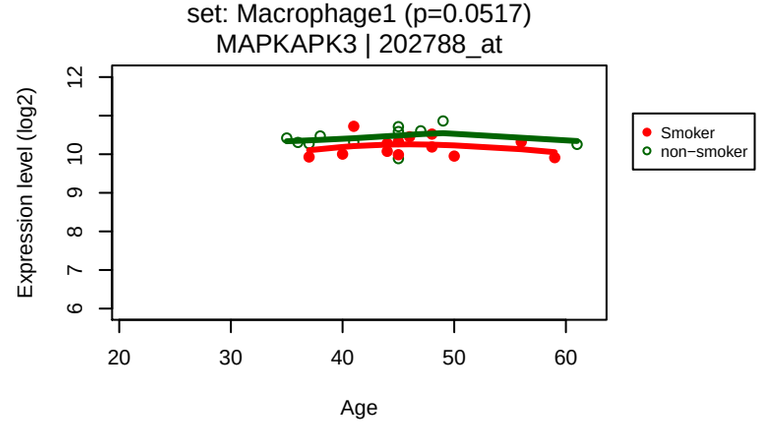

set: SmallAirway1 (p=0.9027)
MAPKAPK3 | 202788_at
Expression level (log2)
Age
Smoker
non−smoker

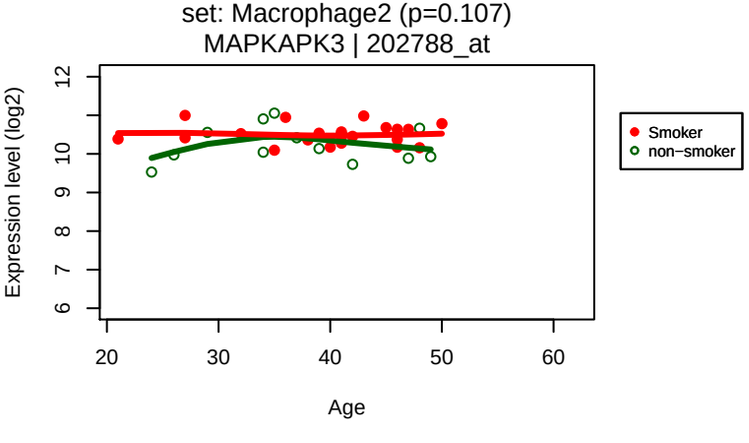

set: SmallAirway2 (p=0.6823)
MAPKAPK3 | 202788_at
Expression level (log2)
Age
Smoker
non−smoker

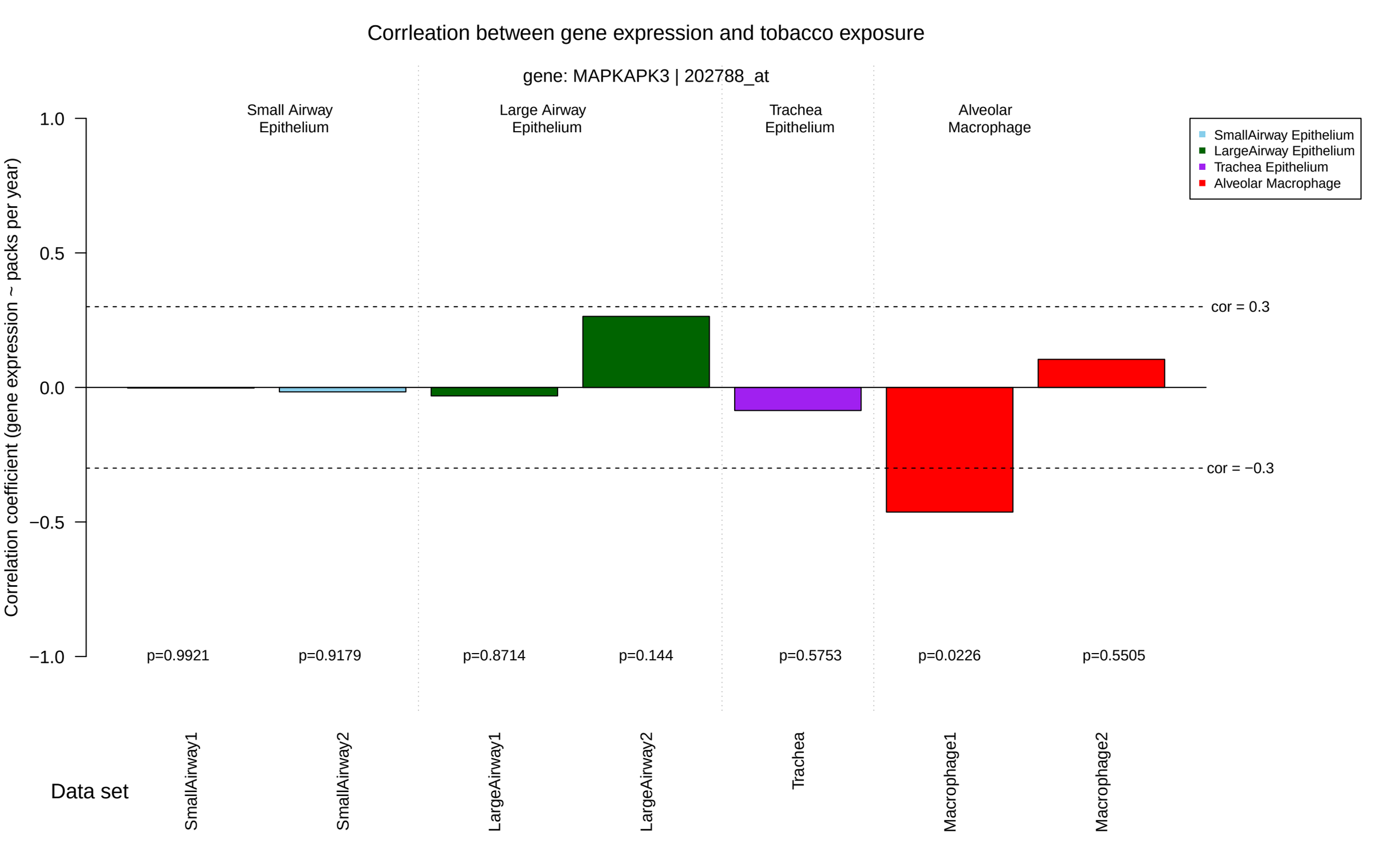

set: LargeAirway1 (p=0.6845)
MAPKAPK3 | 202788_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0363)
MAPKAPK3 | 202788_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3455)
MAPKAPK3 | 202788_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0517)
MAPKAPK3 | 202788_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.107)
MAPKAPK3 | 202788_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPKAPK3 | 202788_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9921
p=0.9179
p=0.8714
p=0.144
p=0.5753
p=0.0226
p=0.5505
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2